

SOC HOME LAB REPORT

SIEM & SECURITY MONITORING WITH WAZUH



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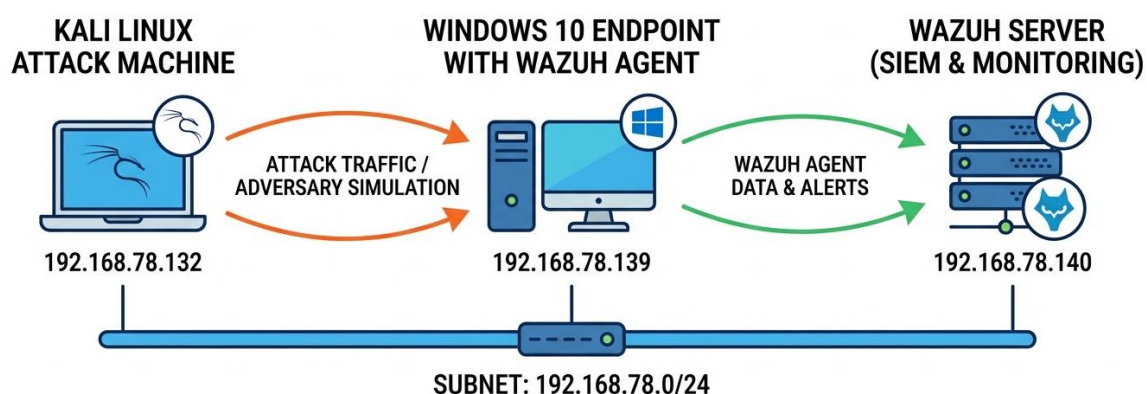
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Executive Summary

This project documents the design, implementation, and evaluation of a hands-on SOC home lab simulating real-world offensive and defensive operations. The lab environment leverages a Kali Linux instance to execute controlled cyber-attacks against a monitored Windows endpoint equipped with a Wazuh Agent and Sysmon.

Lab Architecture & Network Topology



The Adversary (Kali Linux - 192.168.78.132): This node acts as the Red Team or threat actor. It is responsible for generating offensive traffic—such as port scans, brute-force attempts, or payload execution.

The Target (Windows 10 - 192.168.78.139): This is the victim endpoint being actively monitored. Equipped with the Wazuh Agent, it continuously observes internal system state, network connections, and process executions.

The SIEM (Wazuh Server - 192.168.78.140): Acting as the central brain of the defensive operation, this server continuously ingests the encrypted telemetry forwarded by the Windows agent. It parses the raw logs, correlates them against threat intelligence and detection rules, and surfaces actionable alerts for incident response and threat hunting.

Wazuh SIEM Deployment

The Wazuh server (192.168.78.140) utilizes an all-in-one deployment architecture tailored for lab environments. The following core components operate concurrently on the main node:

- **Wazuh Manager:** The central correlation engine that receives agent telemetry in real-time, parses raw logs, and evaluates them against threat detection rules.
- **Wazuh Indexer:** A highly scalable search and analytics engine that securely indexes and stores the parsed security data for high-speed querying.
- **Wazuh Dashboard:** The web-based user interface utilized for real-time data visualization, custom dashboard creation, and incident alert triage.

Threat Detection & Rule Customization

Custom Rule Development: Brute-Force Detection Logic

To effectively detect automated password-guessing attacks originating from the Kali Linux adversary node, custom XML rules were authored within the Wazuh manager. The detection logic relies on a parent-child rule dependency to differentiate between standard user authentication errors and malicious brute-force campaigns (such as those executed via THC-Hydra).

This foundational rule (**ID 100006**) acts as the initial tripwire in the detection pipeline. It is not designed to generate a high-priority alert on its own, but rather to tag specific host telemetry for downstream correlation.

```
<rule id="100006" level="3">
  <field name="win.system.eventID">4625</field>
  <description>Custom Alert: Windows Failed Logon
  Attempt (Event 4625)</description>
</rule>
```

- **Trigger Condition:** The rule parses Windows Security logs and triggers when `win.system.eventID == 4625` is detected, which is the native Microsoft Event ID for a failed logon attempt.
- **Severity Level 3:** Assigned a low severity, this rule operates quietly in the background. This prevents the SOC dashboard from being flooded with false positives resulting from simple user typos.

This composite rule (**ID 100005**) aggregates the findings of the base rule to identify hostile intent. It leverages temporal thresholds and IP tracking to confidently detect automated attacks.

```
<rule id="100005" level="12">
  <if_matched_sid>100006</if_matched_sid>
  <same_field>win.eventdata.ipAddress</same_field>

  <description>
    Custom Alert: Hydra Brute Force Attack Detected
    against Windows Target
  </description>

  <mitre>
    <id>T1110</id>
  </mitre>
<group>windows,authentication,brute_force,custom_rules,</
group>
</rule>
```

- **Rule Dependency (if_matched_sid):** The engine only evaluates this rule if the base failed logon rule (100006) has already been triggered.
- **Threshold Mechanics (frequency="3" timeframe="60"):** The rule logic requires a minimum of three distinct failed login events to occur within a strict 60-second window.
- **Severity Level 12:** Categorized as a high-severity alert, this guarantees immediate visibility for incident responders and is severe enough to automatically trigger an Active Response mitigation script.
- **T1110.001 (Password Guessing):** The rule is explicitly mapped to this specific MITRE sub-technique. This enriches the alert metadata, providing SOC analysts with immediate tactical context regarding the adversary's objective during the alert triage phase.

Custom Rule Development: Reconnaissance (Nmap) Detection Logic

Before an adversary attempts exploitation or credential access, they typically conduct network scanning to identify active hosts, open ports, and vulnerable services. To detect this early-stage reconnaissance from the Kali Linux node, a custom rule was engineered to identify Nmap scanning activity targeting the Windows endpoint.

This rule (**ID 100010**) is designed to detect the distinct network footprint left by port scanners interacting with the Windows Filtering Platform (WFP).

```
<rule id="100010" level="12">
  <if_sid>60103</if_sid>
  <field name="win.system.eventID">5156</field>
  <description>Custom Alert: Possible Nmap Network Scan
Detected</description>
  <mitre>
    <id>T1046</id>
  </mitre>

<group>windows,network,reconnaissance,nmap,custom_rules,
</group>
</rule>
```

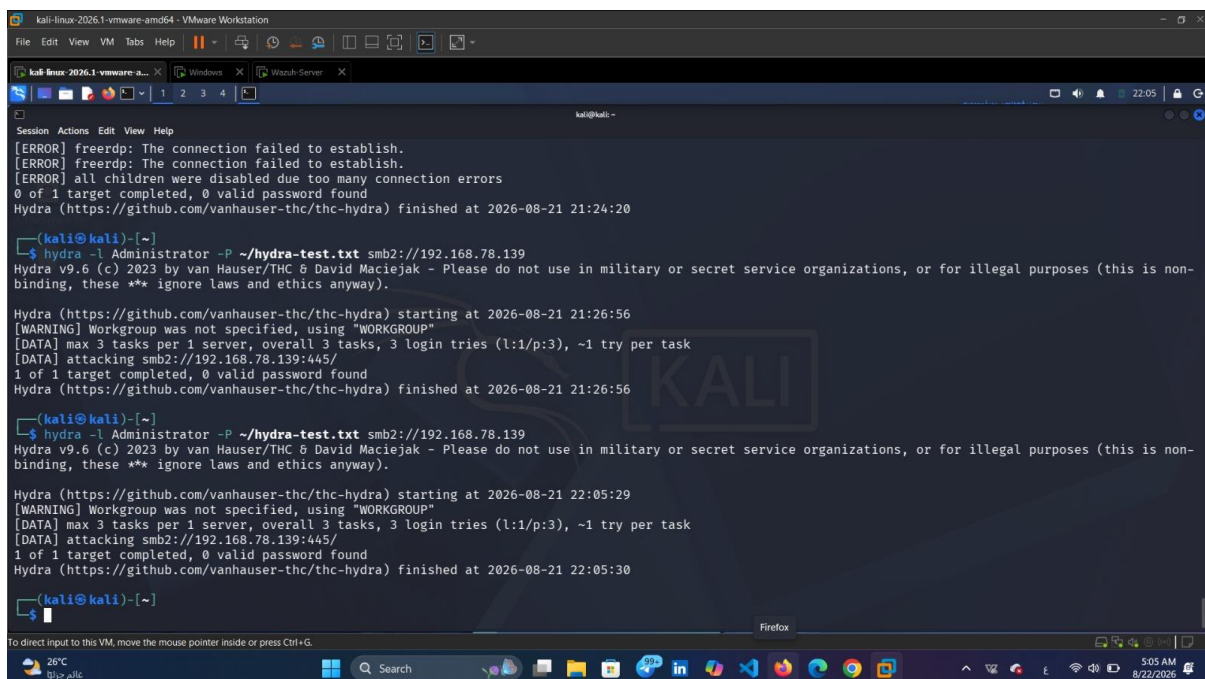
- **Rule Dependency (if_sid):** The logic evaluates this rule only if baseline rule 60103 is triggered, which corresponds to fundamental Windows network connection events.
- **Event Filtering (win.system.eventID):** The engine specifically looks for **Event ID 5156** (The Windows Filtering Platform has allowed a connection). When an attacker runs an Nmap scan, it rapidly initiates connections across hundreds of ports, generating a massive spike in 5156 (Allowed) and 5157 (Blocked) events on the target host.
- **Severity Level 12:** Scanning is a direct precursor to an attack. Assigning a critical severity ensures the SOC is immediately alerted that an adversary is actively mapping the environment's attack surface.
- **Taxonomy & Grouping (group):** The rule is tagged with specific keywords (windows, network, reconnaissance, nmap). This strict categorization is crucial for SOC analysts, allowing them to quickly filter dashboard visualizations and isolate reconnaissance behavior from other alert types.

- **T1046 (Network Service Discovery):** The rule is mapped to the standard MITRE technique for adversarial service enumeration.

Attack Simulation & Threat Hunting

Attack Simulation 1: SMB Brute-Force Execution

To validate the custom detection logic engineered in Wazuh, a controlled brute-force attack was executed from the Kali Linux adversary node against the monitored Windows 10 endpoint.



```
kali@kali:~$ hydra -l Administrator -P ~/hydra-test.txt smb2://192.168.78.139
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-21 21:26:56
[WARNING] Workgroup was not specified, using "WORKGROUP"
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1 try per task
[DATA] attacking smb2://192.168.78.139:445/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-21 21:26:56

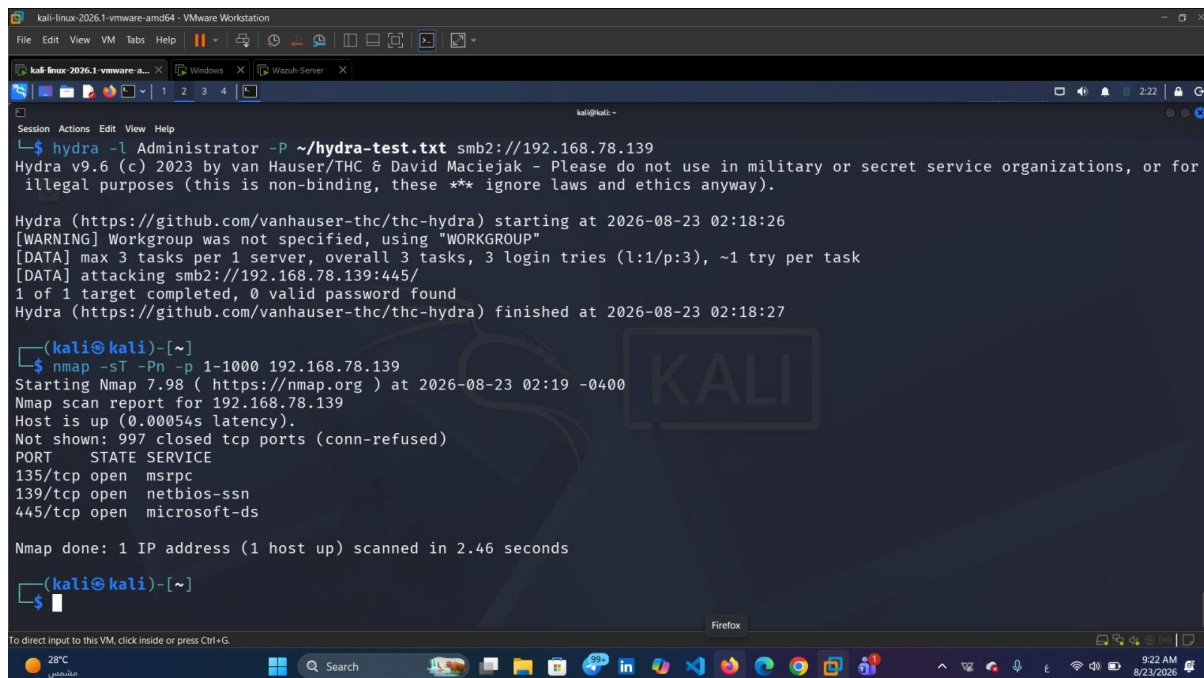
kali@kali:~$ hydra -l Administrator -P ~/hydra-test.txt smb2://192.168.78.139
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-21 22:05:29
[WARNING] Workgroup was not specified, using "WORKGROUP"
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1 try per task
[DATA] attacking smb2://192.168.78.139:445/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-21 22:05:30
```

This specific, high-frequency failure pattern is exactly what is required to trigger the **Level 12 Custom Rule (ID 100005)** configured in the SIEM, successfully testing the environment's ability to detect lateral movement and credential access attempts (MITRE ATT&CK T1110.001).

Attack Simulation 2: Basic Nmap Execution

To validate the custom network scanning detection logic, the Kali Linux adversary node was utilized to conduct active reconnaissance against the target Windows endpoint.



```
kali@kali:~$ hydra -l Administrator -P ~/hydra-test.txt smb2://192.168.78.139
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for
illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-23 02:18:26
[WARNING] Workgroup was not specified, using "WORKGROUP"
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1 try per task
[DATA] attacking smb2://192.168.78.139:445/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-23 02:18:27

(kali@kali)-[~]
$ nmap -sT -Pn -p 1-1000 192.168.78.139
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-23 02:19 -0400
Nmap scan report for 192.168.78.139
Host is up (0.00054s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.46 seconds

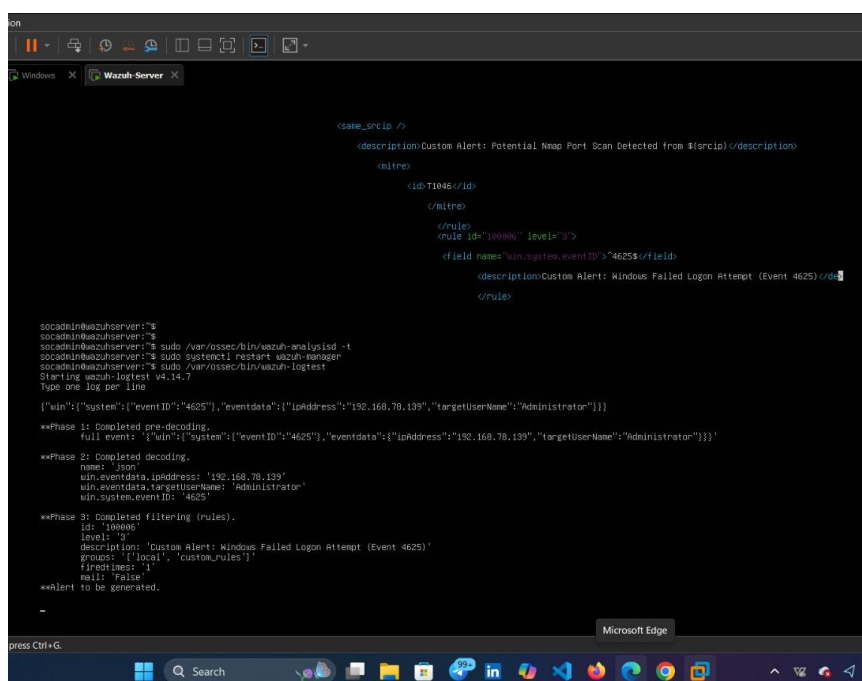
(kali@kali)-[~]
$
```

This specific high-volume, multi-port log generation is exactly what the newly configured **Level 12 Custom Rule (ID 100010)** is designed to ingest, parse, and flag as a MITRE ATT&CK T1046 (Network Service Discovery) violation.

Dashboard Visualization & Incident Triage

Brute-Force Detection Validation

Before analyzing live telemetry, the custom XML rules were verified using the wazuh-logtest utility directly on the SIEM server.



```
<!-- Rule 100006 -->
<!-- Description: Custom Alert: Potential Nmap Port Scan Detected from $(srcip) -->
<!-- MITRE -->
<!-- ID: T1046 -->
<!-- Rule ID: 100006 -->
<!-- Rule Level: 3 -->
<!-- Rule Name: win.system.eventID -->
<!-- Rule Description: Custom Alert: Windows Failed Logon Attempt (Event 4625) -->
</rule>

socadm@wazuhserver:~$ sudo /var/ossec/bin/wazuh-logtest -t
socadm@wazuhserver:~$ sudo systemctl restart wazuh-manager
socadm@wazuhserver:~$ sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.14.2
Type one log per line

{"win":{"system":{"eventID":"4625"},"eventdata":{"ipaddress":"192.168.78.139","targetUserName":"Administrator"}}}

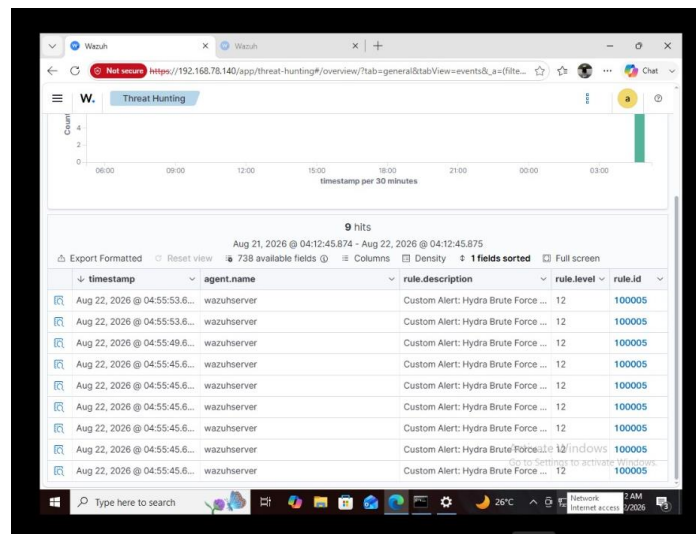
**Phase 1: Completed pre-decoding.
Full event: {'win':{'system':{'eventID':'4625"},"eventdata":{"ipaddress":"192.168.78.139","targetUserName":"Administrator"}}}}

**Phase 2: Completed decoding.
name: 'json'
win.eventdata.ipaddress: '192.168.78.139'
win.eventdata.targetUserName: 'Administrator'
win.system.eventID: '4625'

**Phase 3: Completed filtering (rules).
id: '100006'
level: '3'
description: 'Custom Alert: Windows Failed Logon Attempt (Event 4625)'
groups: ['local', 'custom_rules']
frequency: '1'
mail: 'False'

**Alert to be generated.
```

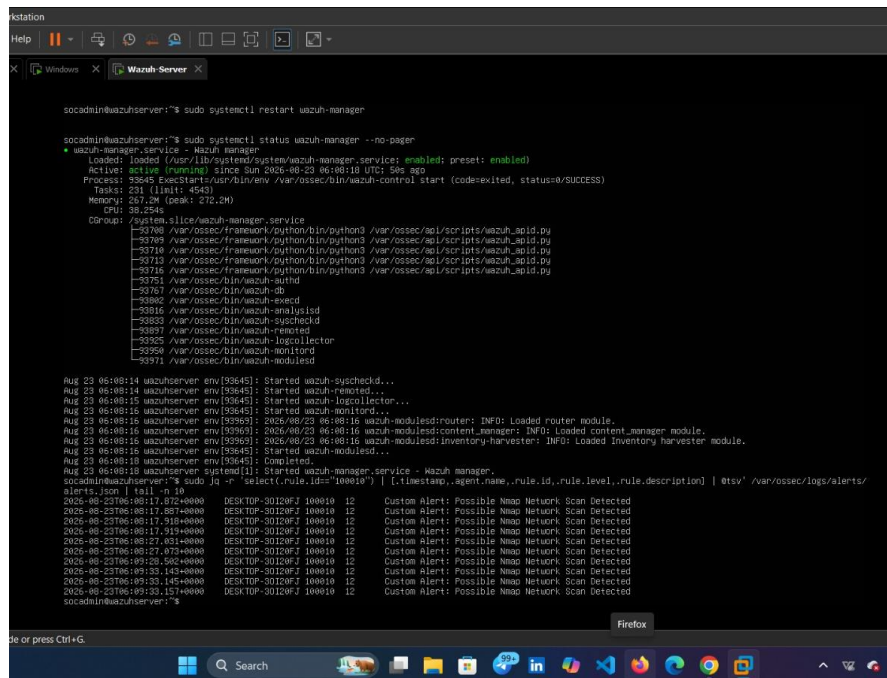
Base Rule Trigger: By feeding a raw JSON log simulating a Windows Failed Logon (Event ID 4625), the engine successfully completed the decoding phase and triggered the baseline **Rule 100006 (Level 3)**.



- Navigating to the **Threat Hunting** module, analysts can visually correlate the attack timeline.
- Querying the dashboard for the specific alert (rule.id: 100005) reveals a dense cluster of hits corresponding to the exact timeframe of the Hydra execution.

Basic Nmap Detection Validation

Before monitoring live traffic, the Wazuh environment was properly initialized to ingest the new detection logic.



```
socadmi@wazuh-server:~$ sudo systemctl restart wazuh-manager

socadmi@wazuh-server:~$ sudo systemctl status wazuh-manager --no-pager
● wazuh-manager.service - Wazuh Manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-23 06:08:18 UTC; 56s ago
     Process: 93549 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Main PID: 231 (limit: 4543)
      Tasks: 267.2M (peak: 272.2M)
         CPU: 39.254s
    CGroup: /system.slice/wazuh-manager.service
            └─ 93700 /var/ossec/framework/python/bin/python3 /var/ossec/apl/scripts/wazuh_apid.py
               93709 /var/ossec/framework/python/bin/python3 /var/ossec/apl/scripts/wazuh_apid.py
               93710 /var/ossec/framework/python/bin/python3 /var/ossec/apl/scripts/wazuh_apid.py
               93712 /var/ossec/framework/python/bin/python3 /var/ossec/apl/scripts/wazuh_apid.py
               93716 /var/ossec/framework/python/bin/python3 /var/ossec/apl/scripts/wazuh_apid.py
               93751 /var/ossec/bin/wazuh-authd
               93767 /var/ossec/bin/wazuh-db
               93802 /var/ossec/bin/wazuh-execd
               93816 /var/ossec/bin/wazuh-enclis
               93833 /var/ossec/bin/wazuh-syscheckd
               93897 /var/ossec/bin/wazuh-repoted
               93925 /var/ossec/bin/wazuh-logcollector
               93950 /var/ossec/bin/wazuh-monitord
               93971 /var/ossec/bin/wazuh-modulesd

Aug 23 06:08:14 wazuhserver env[93645]: Started wazuh-syscheckd...
Aug 23 06:08:14 wazuhserver env[93645]: Started wazuh-repoted...
Aug 23 06:08:15 wazuhserver env[93645]: Started wazuh-logcollector...
Aug 23 06:08:15 wazuhserver env[93645]: Started wazuh-monitord...
Aug 23 06:08:16 wazuhserver env[93663]: 2026/08/23 06:08:16 wazuh-modulesd:router: INFO: Loaded router module.
Aug 23 06:08:16 wazuhserver env[93663]: 2026/08/23 06:08:16 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Aug 23 06:08:16 wazuhserver env[93663]: 2026/08/23 06:08:16 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester module.
Aug 23 06:08:16 wazuhserver env[93645]: Started wazuh-modulesd...
Aug 23 06:08:18 wazuhserver env[93645]: Completed.
Aug 23 06:08:18 wazuhserver systemd[1]: Started wazuh-manager.service - Wazuh manager.
socadmi@wazuh-server:~$ sudo jq -r 'select(.rule.id=="100010") | [.timestamp,.agent.name,.rule.id,.rule.level,.rule.description] | @tsv' /var/ossec/logs/alerts/alerts.json | tail -n 10
2026-08-23T06:08:17.872+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:08:17.887+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:08:17.916+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:08:17.931+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:08:27.631+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:08:27.673+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:09:25.582+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:09:33.145+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:09:33.145+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
2026-08-23T06:09:33.157+0000 DESKTOP-30120F7 100010 12 Custom Alert: Possible Nmap Network Scan Detected
socadmi@wazuh-server:~$
```

- **Service Initialization:** As demonstrated in the terminal capture, the wazuh-manager service was restarted (systemctl restart wazuh-manager) to compile and load the updated XML rule set into the active correlation engine.
- **Logtest Verification:** The wazuh-logtest utility was utilized to simulate the ingestion of **Event ID 5156** (The Windows Filtering Platform has permitted a connection). The engine successfully completed the pre-decoding, decoding, and filtering phases, correctly parsing critical metadata such as the Source Address (Kali IP) and Destination Port

